MD. Hosain Mahmud

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Professional Summary

Software Engineer with experience building backend applications using Node.js, Express.js, and Python-based APIs. Strong foundation in software development, REST APIs, databases, and Linux environments. Familiar with Docker, Git, and FastAPI, with additional research experience in machine learning .

Technical Skills

- **Languages:** C/C++, Python, Go, JavaScript.
- **backend:** Node.js, Express.js, FastAPI, REST APIs
- **Databases:** MySQL, PostgreSQL, MongoDB, SQLite.
- **Tools & Technologies:** Git, GitHub, Docker, Linux/WSL, GitHub Actions, Postman, Azure.
- **Machine Learning:** Transfer learning, CNN, Image Classification, NLP.

Education

Shanto Mariam University of Creative Technology, BSc in CSE. Dec 2020 – Jul 2025

- GPA: 3.82/4.0
- **Relevant Coursework:** OOP, DSA, Operating System, Data Mining, Machine Learning

Projects

Tourism Booking Platform | Node.js • Express.js • MongoDB • Stripe https://github.com/Hosain199/web

- Designed and developed a backend-focused web application for tour and guide booking.
- Implemented RESTful APIs and integrated Stripe payment processing.
- Created database models and user management features.
- Applied MVC architecture and version control using Git.

Job Board API | FastAPI, SQLAlchemy, JWT github.com/Hosain199/JobApplication

- Designed and developed a backend RESTful API for a job board platform with employer and candidate roles.
- Implemented JWT-based authentication and role-based access control for job postings and applications.
- Created database models and migrations using SQLAlchemy 2.0 and Alembic.
- Applied layered architecture (routers, schemas, models) and wrote a 20-test pytest suite.
- Containerized the application with Docker and set up CI with GitHub Actions.

Thesis Projects

Research Project: Freshwater Macro-invertebrates Detection Using deep Learning Conducted research on applying deep learning techniques for image recognition. Developed a pipeline for model training, testing, and evaluation to contribute to advancements in ecological monitoring.

Achievements & Leadership

- **Hackathon Runner-up**: 2nd among 50 teams.
- **Math Olympiad 23**: ranked 25th among 162 participants.
- **ICPC Preliminary Round**: competitive programming participant.
- **President, BASIS Students' Forum**: Led technical workshops and events.